# MID-SEMESTER EXAMINATION, November-2025
# Practical Robotics Projects with Arduino (CSE4571)

**Program: B.Tech (CSE)** **Semester:7th**
**Full Marks: 30** **Time: 2 Hours**

| Subject/Course Learning Outcome | *Taxonomy Level | Ques. Nos. | Marks |
|---|---|---|---|
| CO1: Understand the fundamentals of Arduino hardware and software. | L1, L2 | 1(a) & (c), 2(a) | 6 |
| CO2: Interface various sensors and actuators with Arduino. | L2 | 1(b), 2(b), 3(b) | 6 |
| CO3: Apply programming logic to control robotic systems. | L3 | 2(c), 3(c), 5(c) | 6 |
| CO4: Design and build basic to intermediate level robotics projects. | L3, L4 | 4(b), (c) | 4 |
| CO5: Demonstrate problem-solving and debugging skills in robotics. | L4 | 3(a), 4(a) | 4 |
| CO6: Collaborate in teams to plan, execute, and present robotics projects. | L5 | 5(a), (b) | 4 |

*Bloom's taxonomy levels: Remembering (L1), Understanding (L2), Application (L3), Analysis (L4), Evaluation (L5), Creation (L6)

Answer all questions. Each question carries equal mark.

| | | | |
|---|---|---|---|
| 1. | (a) | What is Arduino? How do you power an Arduino board? | 2 |
| | (b) | What are sensors and how are they used in robotics? | 2 |
| | (c) | What is the basic structure of an Arduino program? | 2 |
| 2. | (a) | Explain how the onboard LED at pin 13 of Arduino Uno board is useful. | 2 |
| | (b) | What are the key differences between analog input and digital input in Arduino? | 2 |
| | (c) | Explain how analogWrite() can be used to control LED brightness. | 2 |
| 3. | (a) | What happens if the LED is connected directly to the Arduino without a resistor, and how does this relate to pin safety? | 2 |
| | (b) | What is the difference between using a pull-up and a pull-down resistor in a push-button circuit with Arduino? | 2 |



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|---|---|---|---|
| | (c) | Write the Arduino code to blink an LED with a 25% duty cycle and a period of 2 seconds. Which Arduino function controls the output pin state? | 2 |
| 4. | (a) | What does PWM stands for and what is it's use in Arduino? | 2 |
| | (b) | Write down the instructions necessary to use the Serial Monitor to display outputs for different purposes? | 2 |
| | (c) | What advantage does the Serial Plotter offer compared to the Serial Monitor? | 2 |
| 5. | (a) | Why is it necessary to divide the total sound travel time by 2 in ultrasonic distance measurements? | 2 |
| | (b) | Describe the roles of the Trigger and Echo pins in the HC-SR04 ultrasonic sensor and write the formula to calculate distance. | 2 |
| | (c) | Write code to display the measured distance from an ultrasonic sensor on an I2C LCD. | 2 |
| | | *End of Questions* | |

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